Analytical Cume

September 28, 2007

1. Discuss ways in which you can obtain information about the composition of an asteroid without flying a mission to them. (20 points)

2. Discuss ways in which analyses in the laboratory provide information that can help interpret the data obtained for asteroids. (20 points)

3. On the basis of your previous discussions, what analytical instruments would you place on a mission that does not return samples? (20 points)

4. On the basis of your discussion in parts (1) and (2), what samples would you return from an asteroid if you had a mission to do so? (20 points)

5. In addition to content, credit will be given for well-written essays with a beginning, middle and end, demonstrating mastery of thought and imagination, even if not entirely technically accurate. An additional 20 points are reserved for the following:

   a) Use of appropriate Figures and Tables
   b) Use of appropriate calculations and quantitative arguments